ZCAS University

MASTER OF SCIENCE IN INFORMATION TECHNOLOGY

MIT111 DISTRIBUTED DATA MANAGEMENT

FINAL EXAMINATION

4th June 2026 **14:00 – 17:00**

TIME ALLOWED: READING AND WRITING – THREE HOURS AND FIVE MINUTES

INSTRUCTIONS:

1. Section A: Question One in Section A is **compulsory.**
2. Sections B: Answer **TWO** Questions from this section.
3. This examination paper carries a total of **100 marks.**

SECTION A – COMPULSORY

QUESTION ONE

(a) Consider the following relational Database schema and the set of applications that are frequently accessing the relation;

DEPARTMENT (DeptNo, DeptName, School, Phone)

Application 1: Find the department details of SITE school.

Application 2: Find the phone of 'Digital Communication' department.

Required:

(i) Define the set of minterm predicates required to horizontally fragment the given schema according to the requirements and check for correctness of the fragments.

[12 marks]

(ii) Fragment the given schema according to your defined minterm predicates.

[8 marks]

(iii) Check the fragments for disjointness and define the statements required to reconstruct the original relation.

[5 marks]

(b) The company ZCAS Real Estates transfers the monthly salary of its employee Mwine Munshi from their bank account at bank A to his bank account at bank B on May 25th. On the following day the company also transfers the outstanding travel expenses to Mwine Munshi's account. At bank A, a bank transfer requires 6 days, i.e. debit on day one, 4 days processing time, and credit entry on day 6. On May 27th Mwine Munshi transfers the monthly rent for his house to his landlord, which has a bank account at bank C. At bank B a transfer only requires 4 days.

Required:

(i) What is the intertwined schedule for these transactions and what are the intermediate states of the bank accounts?

[6 marks]

(ii) Outline a possible serial schedule, its intermediate states, and decide whether the result of this schedule is equivalent.

[10 marks]

(iii) What would happen if any of the transactions were to fail on the last day?

[5 marks]

(iv) Which of the transaction properties would have been used to resolve the problem in (iii)?

[4 marks]

(Total 50 marks)

SECTION B: Answer ANY TWO Questions

QUESTION TWO

In three separate Schools of ZCAS University, three different database systems are used to store information about students, each in simply structured tables as follows:

Examination office CS (MS Access):

StudentsSoB(**Sno**, name, firstname, sex)

StudentsSoCIT(**Sno**, name, firstname, sex)

StudentsSoSS(**Sno**, name, firstname, sex)

Central Student Administration (MS SQL Server:

Students(Sno, fullname, dayOfBirth, sex, courseOfStudies)

University library (Oracle 10g):

Students(cardID, firstname, lastname, memberSince)

Now, a new application for the Student Union needs to query (only read accesses) all this data.

Required:

(a) Discuss with justification whether this problem should be solved by a Distributed, Parallel, or Federated Database System (DBS).

[15 marks]

(b) Reliability and availability of a "*Centralized Data Allocation*" method is Lowest when compared to a "*Completely Replicated Data Allocation*".

Required:

Explain why this happens and justify your answer with a clear distinction between the two data allocation strategies.

[10 marks]

(Total 25 mark)

QUESTION THREE

A Distributed Database System requires the use of a Distributed Database Management System (DDBMS) and a DDBMS maintains a Global System Catalog.

Required:

Discuss the role of a Global System Catalog in a Distributed Database environment and describe the types of database schema objects that are stored in it.

[25 marks]

QUESTION FOUR

In a network of three nodes the relations A, B and C should be stored on nodes N_1, N_2 and N_3. The relations have the following sizes:

- A is 10 Terabytes
- B is 15 Terabytes
- C is 5 Terabytes

The storage capacities of each node are limited to 25 Terabytes.

Required:

(i) Discuss the concept of data replication and explain the different types of data replication strategies.

[9 marks]

(ii) Give two example allocations of the three tables, one as *allocation without replication* and one as *allocation with replication.*

[10 marks]

(iii) Using appropriate examples, discuss one possible advantage and one possible disadvantage of the *allocation of relations with replication.*

[6 marks]

(Total 25 marks)

END OF CIT5641 Distributed Data Management FINAL EXAM MAY/JUNE 2026